

Data Analyst Job Market Analysis: Required Visuals

Salary, experience, and programming language requirements

Brief introduction. This project uses 400 Data Analyst job postings to explore how median salary estimates relate to minimum experience requirements and requested data languages. Salary values were treated as numeric, and non-numeric or missing experience values were excluded from the scatter plot.

Data note: The uploaded CSV uses "Minimum year of experience" and "Programming language (locked)" for the requested experience and data-language fields. Language values such as "Python, R" and "R, Python" were grouped as "Both".

Years of Experience Analysis

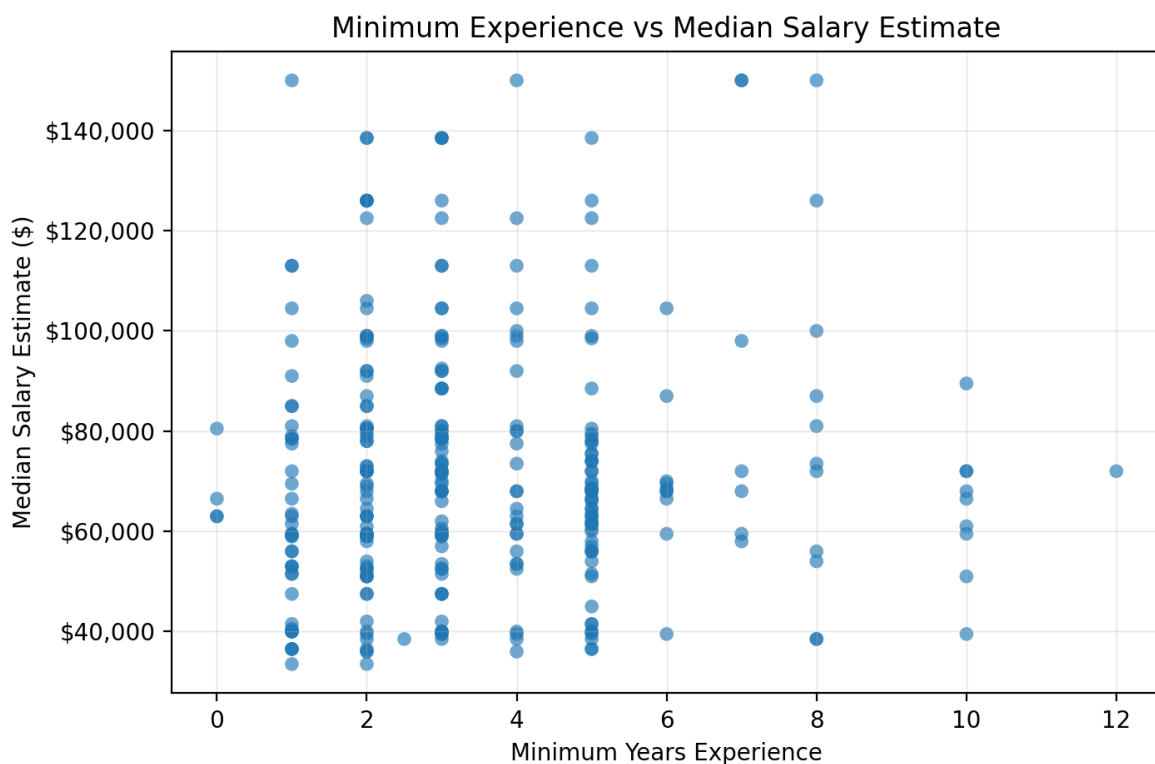


Figure 1. Minimum years of experience vs. median salary estimate.

Interpretation. The scatter plot includes 310 postings with both usable experience and salary values; 90 postings were excluded because the experience field was missing or non-numeric. The overall pattern suggests only a modest positive relationship between minimum years of experience and salary (correlation approximately 0.04). Salaries vary substantially at the same experience level, so experience appears to matter, but it does not fully explain salary differences across postings.

Programming Language Analysis



Figure 2. Median salary estimate grouped by data language requirement.

Interpretation. The language groups are imbalanced, with most postings classified as Neither ($n = 255$), followed by Both ($n = 68$), Python ($n = 62$), and R ($n = 15$). The group medians are Neither: \$68,000; Python: \$69,000; R: \$61,500; Both: \$69,000. The box plot shows that salary distributions overlap across all language categories, although postings asking for both R and Python have a relatively high median in this sample. The small R-only group should be interpreted cautiously.

Reflection

I find it very interesting to be able to integrate the API function to my work. I work very often with large dataset and this is significantly (even from a statistically meaning) increasing the capacities of my work by incorporating the AI in my analyses. I mostly use Excel so it would be interesting to see whether this is possible to integrate the API in there too (I am pretty sure it is possible and AI might even help me to set it up). I think the only limit is the financial one of course cause I understand that the use of the API is not free but I might be able to make a case to my employer to invest in it considering the benefits it would bring.